Deep Learning-Based Depression Detection from Brain MRI

AI-assisted MDD detection using 3D structural MRI and clinical metadata

BY

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Presentation Roadmap

Simple flow from problem to final results



The Problem

Current depression diagnosis is often subjective and hard to standardize

- Grad-CAM-style heatmaps highlight MRI regions that contributed to the prediction.
- Heatmaps are overlaid on MRI slices for visual inspection.
- The goal is to check whether the model focuses on meaningful anatomical regions rather than artifacts.
- Important regions may include hippocampus, prefrontal cortex, and anterior cingulate cortex.

Main challenge

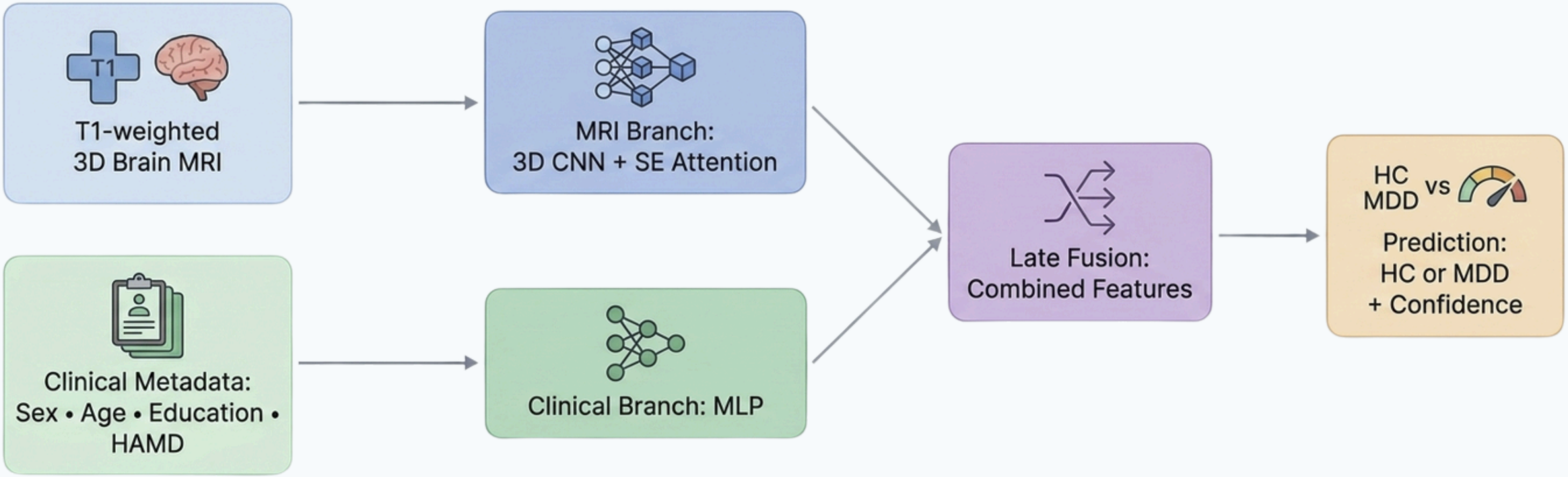
Detect subtle structural patterns related to MDD from full 3D brain MRI volumes.

Important note

The system is a research prototype, not a certified clinical diagnostic tool.

Proposed Solution

A multimodal AI framework for MDD detection



Objectives and Scope

What the project covers and what it does not cover

OBJECTIVES

- Develop an AI-assisted multimodal MDD classifier.
- Preprocess 3D T1 MRI scans and integrate clinical metadata.
- Use healthy-brain autoencoder pretraining as a normative blueprint.
- Evaluate with accuracy, recall, specificity, F1, ROC-AUC, and confusion matrix.
- Support interpretability using Grad-CAM-style heatmaps.

SCOPE

- In scope: T1-weighted structural MRI + clinical metadata.
- Model: 3D CNN, SE attention, MLP branch, late fusion.
- Prototype: upload MRI, enter metadata, show result and heatmap.
- Out of scope: certified diagnosis, treatment planning, fMRI/EEG/PET/CT, other disorders.

Literature Review: Key Gaps

Why this project direction is needed

Traditional ML
Relies on hand-crafted features

2D / Slice Analysis
Can lose 3D spatial relationships

Black Box Models
Hard for clinicians to trust

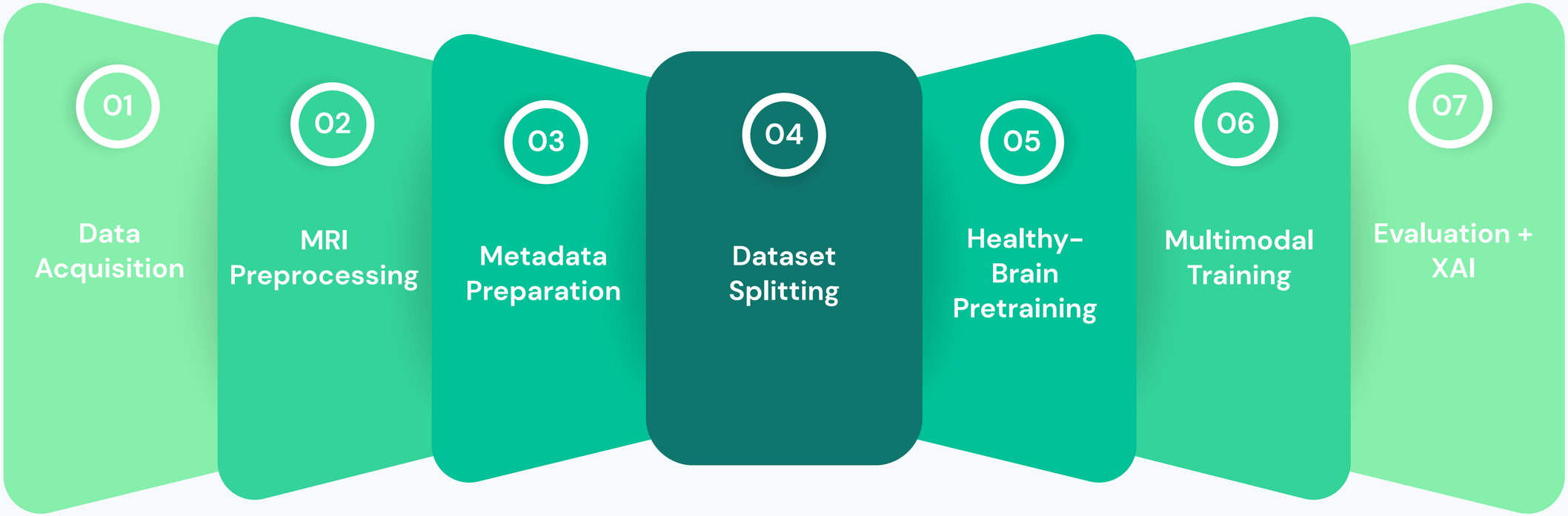
Generalization
Small or single-site data

Project Response

3D volumetric MRI analysis + clinical metadata + autoencoder pretraining + SE attention + explainability support

Methodology Overview

Seven stages from data to explanation



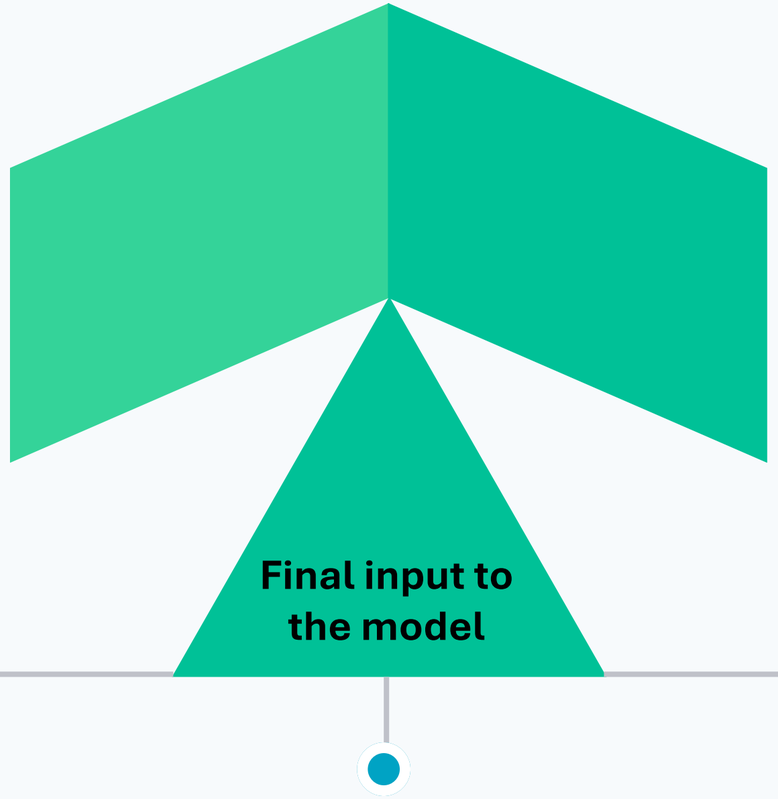
The pipeline keeps the independent test set separate and uses only Healthy Control training samples for the autoencoder pretraining stage.

Dataset and Inputs

Large neuroimaging sources with multimodal inputs

DIRECT Phase II Multi-site MDD and Healthy Control neuroimaging data

REST-meta-MDD
Large MDD research dataset across cohorts



Final input to the model

MRI input
T1-weighted structural MRI
1 × 64 × 64 × 64

Clinical input
- Sex
- Age
- Education
- HAMD

Labels
Healthy Control or MDD

Data Preprocessing

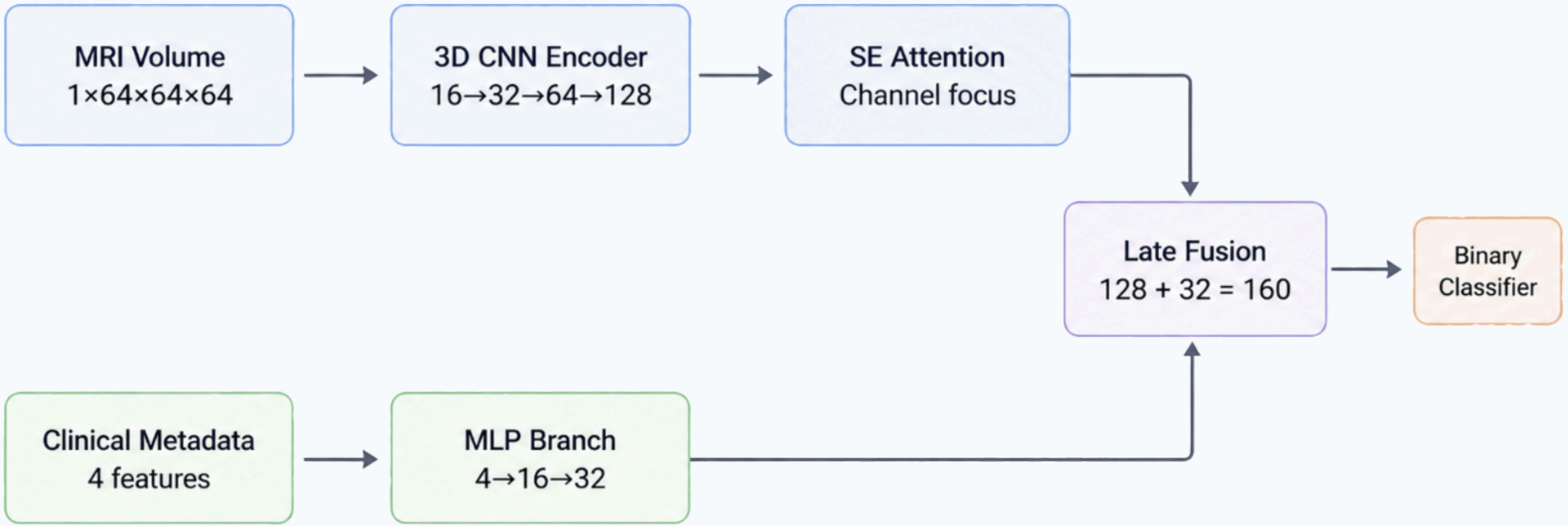
Standardizing MRI and clinical metadata

- Brain extraction and cropping reduce non-brain background.
- MRI volumes are resized to a fixed 3D shape.
- Voxel intensities are normalized using Z-score normalization.
- Clinical values are cleaned, missing values handled, and continuous variables normalized.
- MRI paths, labels, and metadata are fused by subject ID.



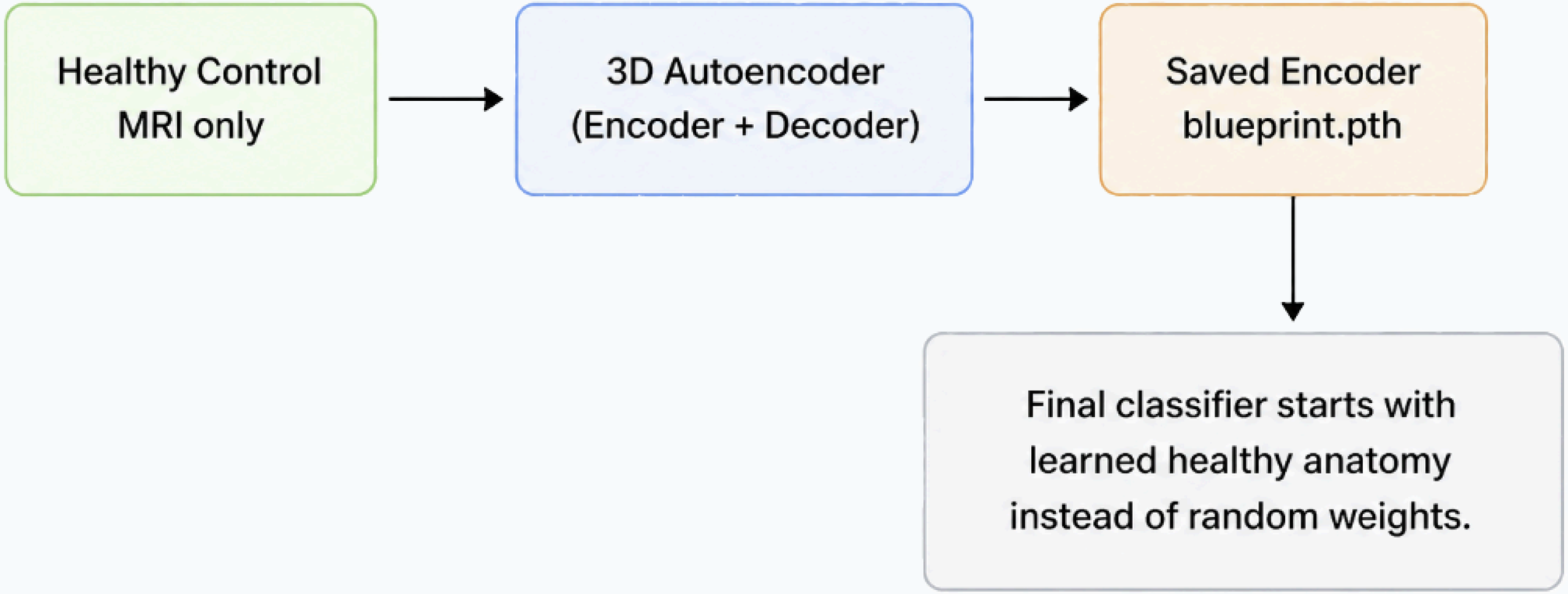
Final Model Architecture

SOTA-inspired multimodal late-fusion network



Normative Anatomy Pretraining

Learning a healthy-brain blueprint before classification

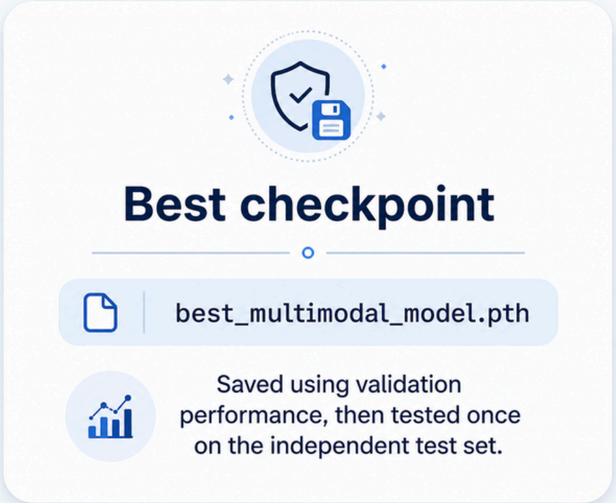


Training evidence: reconstruction MSE decreased from 0.4939 to about 0.1200 after 15 epochs.

Training Strategy

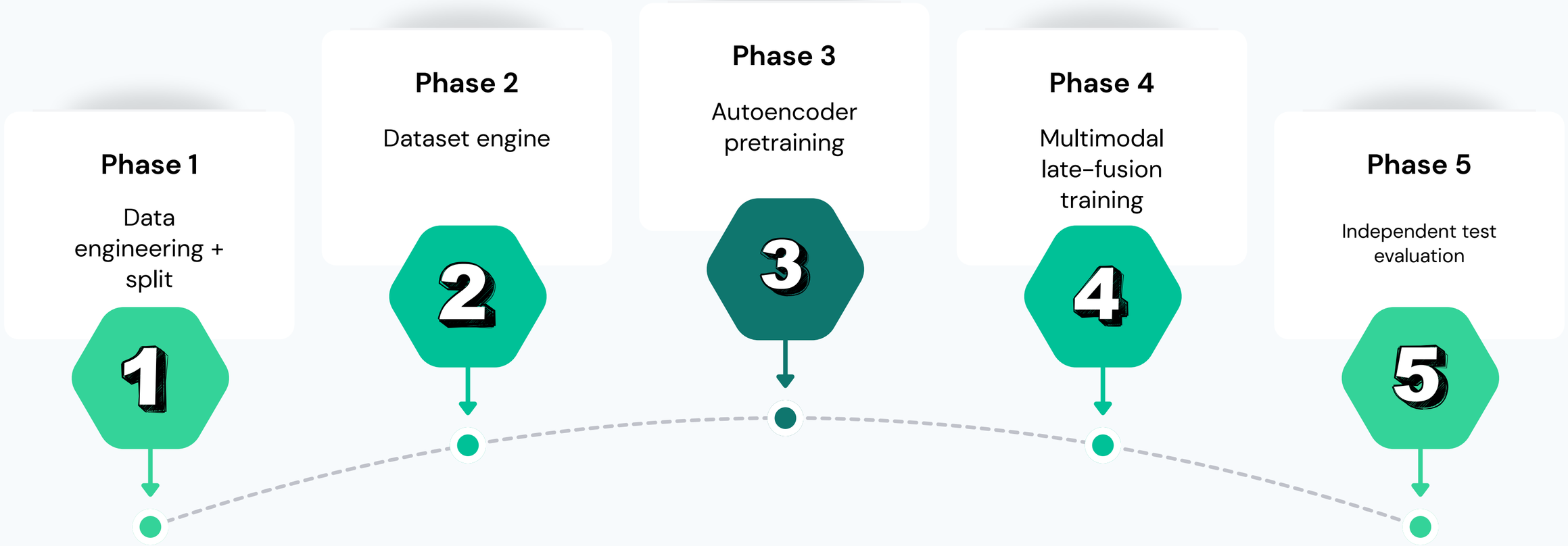
Designed to reduce overfitting and improve robustness

- Unsupervised pretraining: autoencoder trained on Healthy Control scans.
- Supervised training: MRI volumes + clinical metadata for binary classification.
- Loss: sigmoid focal loss to handle difficult / imbalanced samples.
- Regularization: multimodal MixUp, label smoothing, dropout, data augmentation.
- Optimization: AdamW, gradient clipping, cosine annealing warm restarts.



Notebook Implementation Phases

The practical coding workflow behind the model



This structure makes the model development easy to explain:
prepare data → load data → pretrain → train classifier → evaluate.

System Design Requirements

What the prototype must do

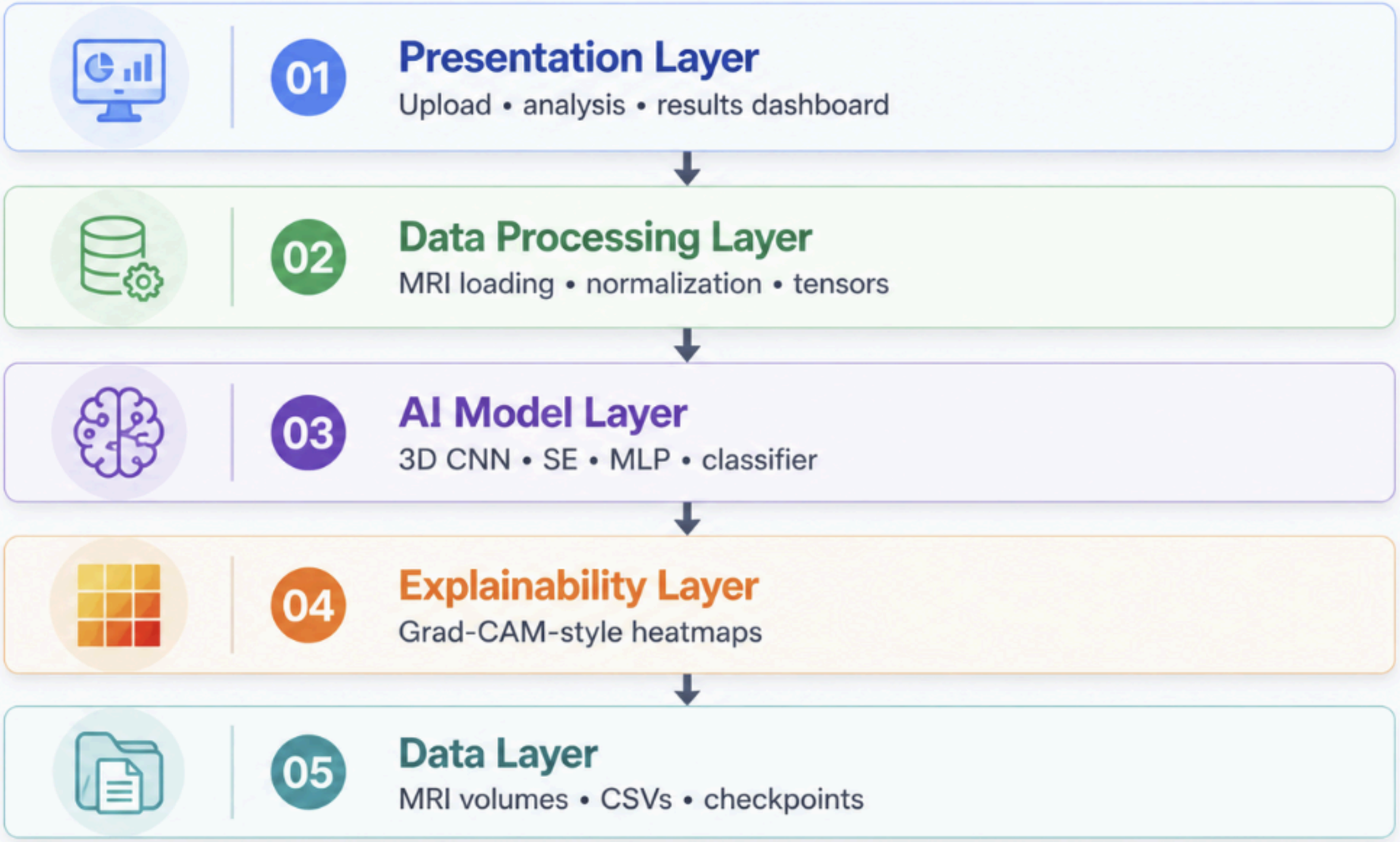
- Upload or select a preprocessed T1-weighted 3D MRI volume.
- Enter or load clinical metadata.
- Run multimodal AI analysis.
- Display predicted class and confidence score.
- Show MRI preview slices and Grad-CAM-style heatmap.
- Allow a new scan workflow and show a research-use disclaimer.

Non-functional focus

- Usability
- Performance
- Reliability
- Security & privacy
- Maintainability
- Ethical use

Layered System Architecture

Frontend, backend, AI model, explainability, and data



System Implementation

A working web prototype connected to an AI backend



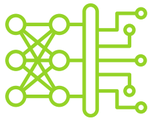
Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS



Backend

- FastAPI
- Python
- Uvicorn
- API endpoint



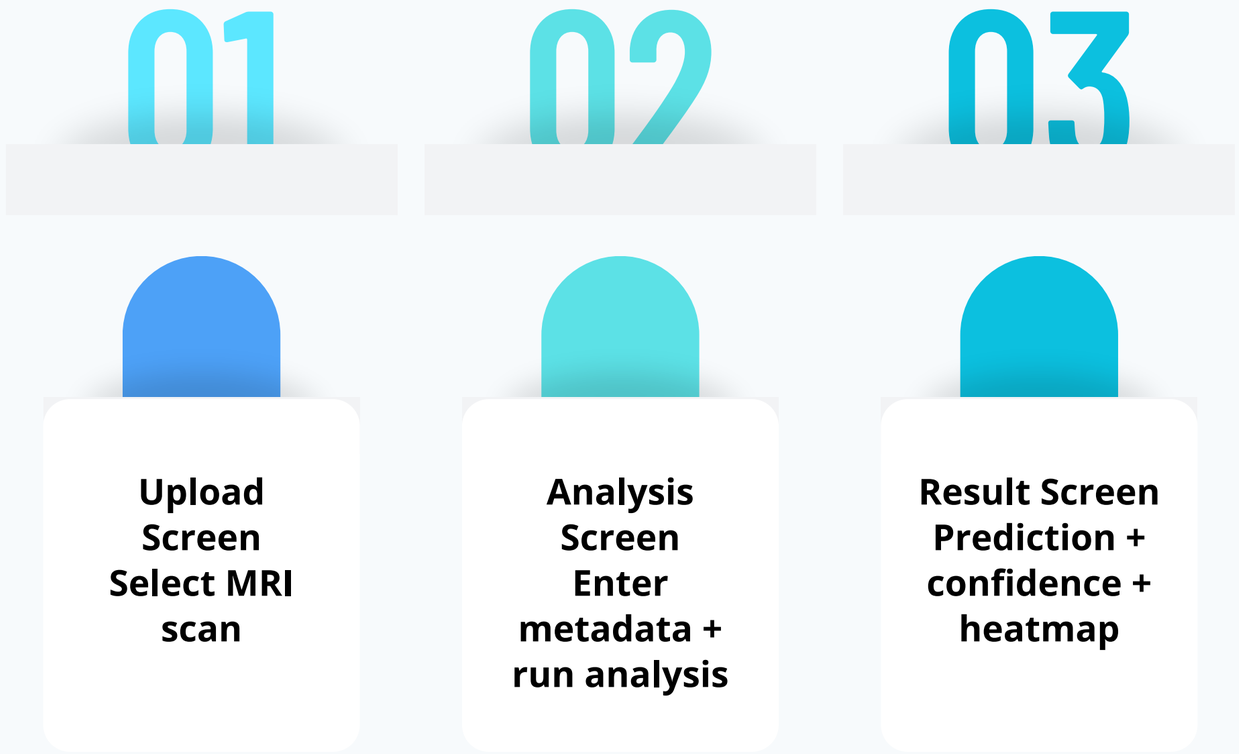
AI Model

- PyTorch
- best_multimodal_model.pth

The frontend sends MRI input and clinical metadata to /api/analyze. The backend returns diagnosis, confidence score, and heatmap output in JSON format.

User Interface Workflow

Three simple screens for the user



- The interface uses a clear medical-style workflow.
- The result screen includes confidence and a disclaimer.
- The user can start a new scan after the analysis.

System Testing

Testing covered frontend, backend, integration, and model performance

- Functional testing: MRI input, metadata, analysis, result display, heatmap, new scan, disclaimer.
- Backend API testing: /api/analyze receives inputs and returns structured JSON.
- Frontend testing: upload, analysis, and result screens verified in browser.
- Integration testing: frontend connected to backend through ngrok public HTTPS URL.
- Model testing: independent test set used only after training and validation.



All listed functional and non-functional tests passed in the prototype environment.

Dataset Split for Testing

Independent test set kept separate

1732

Training samples

217

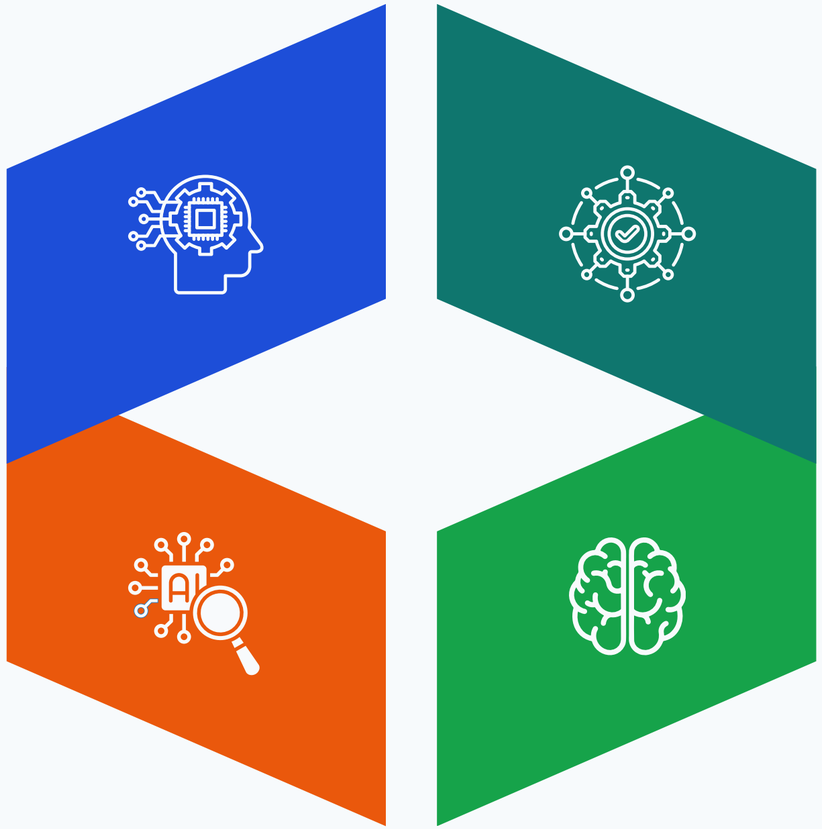
Validation samples

217

Independent test samples

736

Healthy-only pretraining samples



The independent test set was not used during model training or validation, which helps reduce data leakage.

Independent Test-Set Results

Final diagnostic performance of the multimodal model

01

97%

Accuracy

02

0.9870

AUC

03

0.9440

Sensitivity / Recall

04

1.0000

Specificity

05

217

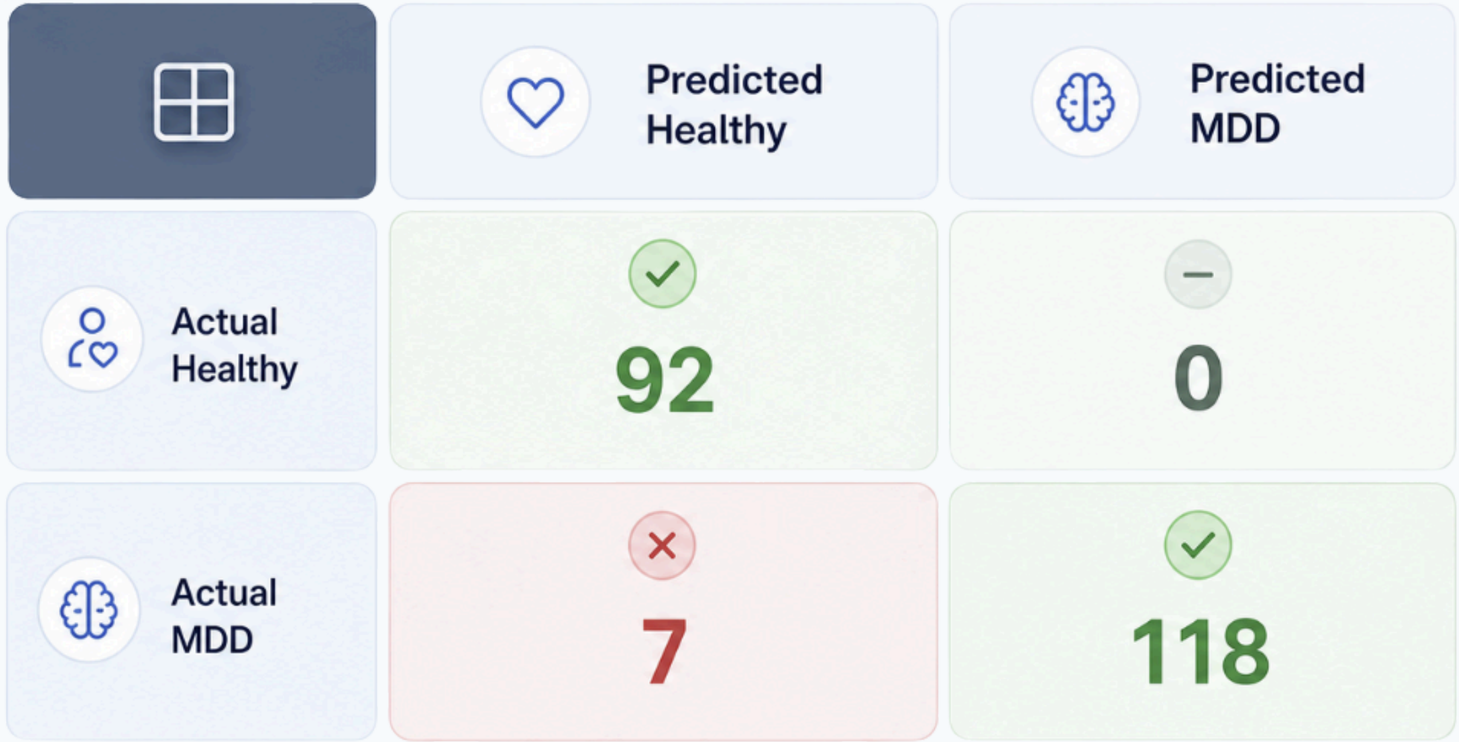
Test samples

Interpretation

The model showed strong research performance on the independent test set, but it still requires external clinical validation before real use.

Confusion Matrix Interpretation

Detailed view of correct and incorrect predictions



Correct: 92 Healthy + 118 MDD. No false positives. False negatives: 7 MDD cases predicted as Healthy, so reducing FN is a key future priority.

Explainable AI: Grad-CAM-Style Heatmaps

Making model decisions easier to inspect

- Grad-CAM-style heatmaps highlight MRI regions that contributed to the prediction.
- Heatmaps are overlaid on MRI slices for visual inspection.
- The goal is to check whether the model focuses on meaningful anatomical regions rather than artifacts.
- Important regions may include hippocampus, prefrontal cortex, and anterior cingulate cortex.



Challenges and Limitations

What must be improved before clinical use

- MRI files are large and require careful preprocessing and storage handling.
- Frontend and backend integration needs stable API deployment beyond ngrok.
- The model requires external validation on new hospitals, scanners, and populations.
- Grad-CAM heatmaps need review by medical experts.
- The prototype is not a certified medical diagnostic system.

Most important limitation

Strong test results do not equal clinical approval. Real deployment requires expert review, privacy controls, and broader validation.

Future Work

Next steps to improve reliability and clinical readiness

01 Validate on external datasets and local Saudi clinical MRI data.

02 Improve 3D Grad-CAM and use anatomical atlas-based explanations.

03 Reduce false negatives through threshold tuning and cost-sensitive learning.

04 Deploy backend securely with HTTPS, authentication, logging, and safe file handling.

05 Strengthen privacy: anonymization, encryption, access control, and audit logs.

06 Explore more modalities and federated learning for privacy-preserving multi-hospital training.

Final Conclusion

A complete academic AI-assisted MDD detection prototype



01 What was built

- Multimodal dataset pipeline
- 3D MRI + clinical metadata model
- FastAPI backend + web interface
- Prediction, confidence, MRI preview, heatmap

02 What it achieved

- 97% test accuracy
- 0.9870 AUC
- Strong specificity
- End-to-end tested workflow

03 How to present it

- Research decision-support only
- Promising but not clinical certification
- Needs external validation and expert review

Overall, the project demonstrates how 3D MRI, clinical metadata, deep learning, and explainability can be combined into one decision-support framework.

THANK YOU

Questions?

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